Module-7 Network Fundamental

Section 1: Multiple Choice Questions

Q1. Which of the following messages in the DHCP process are broadcasted? (Choose two)

a) Request
b) Offer
c) Discover
d) Acknowledge

ANS: a) Request and c) Discover

Q2. Which command would you use to ensure that an ACL does not block web-based TCP traffic?

a) permit any
b) permit tcp any any eq 80
c) permit tcp any eq 80
d) permit any any eq tcp

ANS: b) permit tcp any any eq 80

Section 2:

Q1. Explain network topologies.

ANS: Network topology refers to the physical or logical arrangement of computers and devices in a network — basically, how everything is connected and how data flows.

1. **Bus Topology**

All devices are connected to a single central cable called a "bus."

- **How it works:** Data travels in both directions along the cable.
- **Advantages:** Simple and inexpensive, uses less cable.
- **Disadvantages:** If the main cable fails, the whole network goes down. Performance slows with more devices.

2. **Star Topology**

All devices are connected to a central device like a hub or switch.

- **How it works:** Data passes through the central hub before reaching the destination.
- **Advantages:** Easy to install and manage. Failure of one device does not affect others.
- **Disadvantages:** If the central hub fails, the network stops. Requires more cable.

3. **Ring Topology**

Devices are connected in a circular loop.

- **How it works:** Data travels in one direction (or sometimes both) around the ring.
- **Advantages:** Organized data transmission. No data collisions.
- **Disadvantages:** Failure of one device can disrupt the network. Harder to troubleshoot.

4. **Mesh Topology**

Every device is connected to every other device.

- **How it works:** Data can take multiple paths to reach its destination.
- **Advantages:** Very reliable and fault-tolerant. High performance.
- **Disadvantages:** Expensive and complex. Requires lots of cabling.

5. **Tree Topology**

A combination of star and bus topologies, arranged in a hierarchical structure.

- **How it works:** Multiple star networks are connected to a main bus.
- **Advantages:** Scalable and easy to expand. Structured and manageable.
- **Disadvantages:** If the backbone fails, many nodes are affected. Complex setup.

6. **Hybrid Topology**

A mix of two or more different topologies.

- **How it works:** Combines features of multiple topologies (e.g., star + mesh).
- **Advantages:** Flexible and customizable. Can optimize performance.
- **Disadvantages:** Complex design. Expensive.

2. TCP/IP Networking Model

The TCP/IP model is a 4-layer framework that describes how data is transmitted across networks. It is the practical model used in all modern networking, including the Internet.

The Four Layers

- **Network Access Layer:** Handles physical transmission of data. Deals with MAC addresses, cables, and hardware. (Ethernet, Wi-Fi)
- **Internet Layer:** Handles logical addressing and routing using IP addresses. Key protocols: IP, ICMP. Unit of data: Packet.

- **Transport Layer:** Manages end-to-end communication. TCP = reliable, connection-oriented. UDP = fast, connectionless.
- **Application Layer:** Interface for user applications. Protocols: HTTP, FTP, DNS, DHCP, SMTP, SSH, Telnet.

Data is encapsulated (headers added) as it moves down the layers and decapsulated (headers removed) as it moves up on the receiving end.

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3. LAN and WAN Networks

LAN – Local Area Network

A network covering a small geographic area such as a home, office, or building. Privately owned, high speed (100 Mbps to 10 Gbps), low latency, and low cost. Uses Ethernet (Cat5e/Cat6) or Wi-Fi.

WAN – Wide Area Network

A network spanning large geographic areas such as cities or countries. Owned by ISPs or telecom providers. Lower speed and higher latency than LAN. Uses leased lines, MPLS, fiber, DSL, and satellite. The Internet is the largest WAN.

Key Differences

- **Scope:** LAN = local area. WAN = wide/global area.
- **Ownership:** LAN = private. WAN = leased from ISP.
- **Speed:** LAN = faster. WAN = relatively slower.
- **Cost:** LAN = cheap. WAN = expensive.

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4. Operation of a Switch

A switch is a Layer 2 device that connects devices within a LAN. It forwards data only to the correct destination port using MAC addresses — unlike a hub which sends to all ports.

How it Works

- **MAC Learning:** The switch reads the source MAC address of incoming frames and records it in its MAC address table (CAM table) along with the port number.
- **Known Unicast:** If the destination MAC is in the table, the frame is sent only to that specific port.
- **Unknown Unicast:** If the destination MAC is not in the table, the frame is flooded to all ports except the source port.
- **Broadcast:** Frames with destination FF:FF:FF:FF:FF:FF are sent to all ports.
- **Full-Duplex:** Each switch port supports simultaneous send and receive, eliminating collisions.

- **VLANs:** Managed switches use VLANs to logically segment the network into separate broadcast domains.

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5. Network Devices – Purpose and Functions

- **Hub:** Layer 1. Broadcasts data to all ports. No intelligence. Obsolete.
- **Switch:** Layer 2. Forwards frames using MAC addresses. Creates separate collision domains per port.
- **Router:** Layer 3. Routes packets between different networks using IP addresses. Connects LAN to WAN/Internet. Provides NAT and DHCP.
- **Firewall:** Monitors and filters network traffic based on security rules. Protects the internal network from external threats.
- **Access Point (AP):** Allows wireless devices to connect to a wired network via Wi-Fi (IEEE 802.11).
- **Modem:** Converts digital signals to analog (and vice versa) for transmission over ISP lines (DSL, cable, fiber).
- **Repeater:** Layer 1. Regenerates/amplifies signals to extend network range.
- **Bridge:** Layer 2. Connects two network segments using MAC addresses to filter traffic.
- **Proxy Server:** Acts as an intermediary between clients and the Internet. Used for caching, filtering, and security.

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6. Media, Cables, Ports, and Connectors

Cables

- **Cat5e:** Twisted pair copper. Up to 1 Gbps, 100 m. Standard LAN cable.
- **Cat6:** Twisted pair copper. Up to 10 Gbps (55 m) / 1 Gbps (100 m). Reduced crosstalk.
- **Cat6a:** Twisted pair copper. Up to 10 Gbps over full 100 m.
- **Single-Mode Fiber (SMF):** Laser light. Long distances (km range). Used in WANs and backbones.
- **Multi-Mode Fiber (MMF):** LED light. Shorter distances (up to 550 m). Used in data centers.
- **Coaxial Cable:** Used for broadband Internet (cable modem) and older Ethernet.

Connectors

- **RJ-45:** Standard 8-pin connector for Ethernet (twisted pair) cables.
- **RJ-11:** Telephone/DSL connector. Smaller than RJ-45.
- **LC:** Small fiber optic connector. Common in modern data centers.
- **SC:** Push-pull fiber optic connector. Used in older installations.
- **BNC:** Used with coaxial cables in older networks and CCTV.
- **F-Type:** Screw-on coaxial connector for cable TV and broadband.

Ports

- **Ethernet (RJ-45) Port:** Standard wired network port on computers, switches, and routers.
- **SFP Port:** Modular port accepting hot-swap transceivers for fiber or copper uplinks.
- **Console Port:** Management port (RJ-45 or USB) for configuring Cisco devices.

Cable Wiring Types

- **Straight-Through:** Same pinout both ends. Connects different device types (PC to switch, switch to router).
- **Crossover:** Different pinouts each end. Connects same device types (PC to PC). Mostly obsolete due to Auto-MDIX.

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7. Network Devices and Hosts

Network Devices

Network devices are hardware components that form the infrastructure of a network. They manage, direct, and forward data between hosts. They do not generate user data — they only move it. Examples: switches, routers, firewalls, access points, hubs, and modems.

Hosts

A host is any end device connected to a network that can send or receive data. Every host has a unique IP address. Hosts are the source or destination of network communication.

Types of Hosts

- **Client:** Requests services from a server (e.g., a laptop browsing the web).
- **Server:** Provides services to clients (e.g., web server, file server, DHCP server).
- **Other hosts:** Smartphones, tablets, printers, IP cameras, VoIP phones, IoT devices.

Key Difference

Network devices move data between hosts (infrastructure). Hosts are the endpoints that generate and consume data. A router is a network device; a laptop is a host.